

Ioannis Markakis

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RESEARCH EXPERIENCE

- **Research Associate** September 2025 - September 2027
University of Cambridge
– Project: “Latent Space Semantics for Dependent Type Theory”
Cambridge, United Kingdom
- **Research Assistant** October 2024 - August 2025
University of Cambridge
– Project: “New Categorical Foundations for Compositional Computer Science”
Cambridge, United Kingdom

EDUCATION

- **PhD in Computer Science** October 2020 - July 2025
University of Cambridge
– Thesis title: “An inductive approach to ω -categories and their computads”
– Supervisor: Prof. Jamie Vicary
– Submission: September 2024
Cambridge, United Kingdom
- **MA in Mathematics** September 2017 - May 2020
University of Maryland
– GPA: 4.0/4.0
College Park, Maryland, United States
- **Ptychion in Mathematics (Bachelor’s degree)** September 2013 - May 2017
University of Crete
– GPA: 9.34/10 “Excellent”
– Class Rank: 1st out of 97 students in the academic year 2016-2017
– Erasmus exchange at the University of Barcelona during the 3rd year of studies
– Completed 6 graduate-level courses
Heraklion, Greece

PUBLICATIONS

- **Articles**
 - [1] T. Benjamin, I. Markakis, W. Offord, C. Sarti, and J. Vicary. *Naturality for higher-dimensional path types*. *Proceedings of the 40th Annual ACM/IEEE Symposium on Logic in Computer Science (LICS 2025)*. 2025.
 - [2] T. Benjamin, I. Markakis, and C. Sarti. *CaTT contexts are finite computads*. *Electronic Notes in Theoretical Informatics and Computer Science*. Vol. 4 - Proceedings of MFPS XL. 2024, 5.
 - [3] C. J. Dean, E. Finster, I. Markakis, D. Reutter, and J. Vicary. *Computads for weak ω -categories as an inductive type*. *Advances in Mathematics* 450 (2024).
 - [4] I. Markakis. *Computads for generalised signatures*. *Journal of Pure and Applied Algebra* 228.9 (2024).
- **Preprints**
 - [5] T. Benjamin and I. Markakis. *Hom ω -Categories of a computad are free*. *Journal of the London Mathematical Society* (2025). To appear.
 - [6] T. Benjamin, I. Markakis, W. Offord, C. Sarti, and J. Vicary. *Higher Eckmann-Hilton Arguments in Type Theory*. Preprint. 2025.
 - [7] T. Benjamin and I. Markakis. *Invertible cells in ω -categories*. Preprint. 2024.
 - [8] D. I. Dais and I. Markakis. *Computing minimal generating systems for some special toric ideals*. Preprint. 2017.
- **Formalisation**
 - [9] I. Markakis. *Agda formalisation of computads for ω -categories*. 2023.

SCIENTIFIC TALKS

- **(Invited) Queer and Trans People in Category Theory 2026**
Title: *Computads with invertible generators*
August 2026
Hamburg, Germany
- **TYPES 2026**
Title: *Computads with invertible generators*
May 2026
Gothenburg, Sweden
- **(Invited) Higher Categories, Polygraphs and Homotopy Seminar (IRIF)**
Title: *Computads with invertible generators*
April 2026
Paris, France
- **Category Theory 2024**
Title: *Opposite and hom weak ω -categories*
April 2024
Santiago, Spain
- **(Invited) Masaryk University Algebra Seminar**
Title: *A novel approach to ω -categories*
April 2024
Brno, Czech Republic
- **107th Peripatetic Seminar on Sheaves and Logic**
Title: *Computads for generalised signatures*
April 2023
Athens, Greece
- **30th Yorkshire and Midlands Category Theory Seminar**
Title: *Computads for generalised signatures*
March 2023
Birmingham, United Kingdom
- **Topology, Algebra and Categories in Logic 2022**
Title: *Computads for weak ω -categories*
June 2022
Coimbra, Portugal

TEACHING EXPERIENCE

- **Lecturer**
University of Cambridge
October 2025 - February 2026
Cambridge, United Kingdom
– Lectured the Part II course “Denotational Semantics” and prepared the exam
- **Supervisor**
University of Cambridge
October 2020 - February 2026
Cambridge, United Kingdom
– Supervised undergraduate computer science students in small groups for the courses “Discrete Mathematics” (Michaelmas 2020, 2021, and 2024), “Logic and Proof” (Lent 2022, 2023, 2024, 2025 and 2026), and “Computation Theory” (Lent 2023)
– Organized lab sessions for the graduate course “Advanced Topics in Category Theory” (Lent 2023, 2025, 2026)
- **Graduate Teaching Assistant**
University of Maryland
September 2017 - May 2020
College Park, Maryland, United States
– Independent instruction for the course “College Algebra and Trigonometry” (Fall 2017)
– Teaching assistance for the courses “Precalculus” (Fall 2018 and 2020) and “Calculus I” (Spring 2018)
– Graded weekly assignments for the undergraduate courses “Introduction to Abstract Algebra” (Spring 2019) and “Axiomatic Set Theory” (Spring 2019) as well as the graduate courses “Logic I” (Fall 2020) and “Algebraic Geometry I” (Fall 2020)
- **Undergraduate Teaching Assistant**
University of Crete
Spring 2014 and 2016
Heraklion, Crete, Greece
– Exercise tutorials for groups of 50 - 100 students for the course “Foundations of Mathematics”

AWARDS AND SCHOLARSHIPS

- **College Scholarships**
Fitzwilliam College
March and November 2022
Cambridge, United Kingdom
– Two scholarships for academic excellence in doctoral studies
- **Departmental Scholarship**
University of Cambridge
Sep 2020
Cambridge, United Kingdom
– Four-year doctoral scholarship from the Department of Computer Science and Technology at the University of Cambridge
- **Graduate Scholarship**
Onassis Foundation
October 2020 - September 2024
Vaduz, Liechtenstein

– Four-year doctoral scholarship for studies at the University of Cambridge

• **“Stylianos Pichorides” Award**

University of Crete

November 2016

Heraklion, Greece

– Award from the Department of Mathematics and Applied Mathematics for excellence in undergraduate studies

• **“Chrysanthos and Anastasia Karydis” Award**

University of Crete

July 2016

Heraklion, Greece

– Award for the highest entrance score in the Department of Mathematics and Applied Mathematics

OTHER ACADEMIC ACTIVITIES

• **Logic in Computer Science 2026**

Reviewer

March 2026

Lisbon, Portugal

– Reviewed papers for the conference

• **Logic and Semantics Seminar**

Organiser

February 2025 -

Cambridge, UK

– Organising the CLASH group’s seminar series and inviting speakers

• **Mathematical Structures in Computer Science**

Reviewer

January 2026

– Reviewed papers for the journal

• **Theory and Applications of Categories**

Reviewer

June 2025

– Reviewed papers for the journal

• **12th Symposium on Compositional Structures**

Program Committee Member

April 2024

Birmingham, United Kingdom

– Reviewed two papers for the conference

• **Logical Methods in Computer Science**

Reviewer

October 2023

– Reviewed papers for the journal

• **28th and 32nd Yorkshire and Midlands Category Theory Seminar**

Workshop Organizer

September 2022 and August 2023

Cambridge, United Kingdom

– Responsible for local organization

• **Applied Category Theory 2021**

Conference Organizer

June 2021

Cambridge, UK

– Responsible for program organization and communication with speakers

– Support for remote participation